



Exercise 5

Data Analytics

Advanced SQL

NULL FUNCTIONS

Table 1 Employees

Question 1

SELECT employee_id, name, IFNULL(salary, 0) AS salary_with_default
FROM Employees;

Output

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Question 2

SELECT employee_id, name, IFNULL(department, 'Not Assigned') AS department_name
FROM Employees;

Output

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2: Orders

Question 3

SELECT order_id, customer_id
FROM orders
WHERE ISNULL(customer_id)

Output

order_id	customer_id
103	NULL



Question 4

```
SELECT order_id, customer_id, IFNULL(delivery_date, 'Pending') AS delivery_status  
FROM Orders;
```

Output

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3: Student

Question 5

```
SELECT student_id, name, IFNULL(grade, 0) AS final_grade  
FROM Students;
```

Output

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Question 6

```
SELECT COUNT(*) AS not_graded_count  
FROM Students  
WHERE grade IS NULL
```

Output

not_graded_count

1



Table 4: Products

Question 7

SELECT product_id, name, ~~price~~
~~price~~ (price - COALESCE(discount, 0)) AS qual-price
FROM Products;

Output

product_id	name	qual-price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Table 5: Customers

Question 8

SELECT COUNT(*) AS missing-email-count
FROM Customers;
WHERE email IS NULL

Output

missing-email-count
2

Question 9

SELECT customer_id, name
COALESCE(email, 'NO Email') AS email-display
FROM Customers;

Output

customer_id	name	email-display
1	Linda	NO Email
2	Joseph	joseph@marb.com
3	Mia	NO Email



Table 6: Payments

Question 10

SELECT payment_id, IFNULL(method, 'Unknown') AS method_display, status
FROM Payment;

Output

payment_id	method_display	status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

Table 7: Inventory

Question 11

~~**SELECT** payment_id, IFNULL(method, 'Unknown') AS method_display, status
FROM Pay;~~

SELECT item_id, item_name, IFNULL(quantity, 0) AS quantity_checked
FROM Inventory;

Output

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8: Employees - Extra

Question 12

SELECT emp_id, COALESCE(bonus, commission) AS first_available_reward
FROM Employees - Extra;

emp_id	first_available_reward
1	300
2	100
3	NULL



Table 9: Classes

Question 13

```
SELECT COUNT(*) AS no_room_count  
FROM Classes  
WHERE room IS NULL;
```

Output

no_room_count
2

Table 10: Attendance

Question 14

```
SELECT student_id, date, IFNULL(status, 'Not Marked') AS attendance_status  
FROM Attendance;
```

Output

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11: Bank Accounts

Question 15

```
SELECT account_id  
IFNULL(account_type, 'Unknown') AS type_display,  
IFNULL(balance, 0) AS balance_checked  
FROM Bank_Accounts;
```

Output

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000



Table 12: Projects

Question 16

SELECT project_id, title, IFNULL(start_date, 'TBD') AS start_display
FROM Projects;

Output

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13: Reviews

Question 17

SELECT review_id, product_id,
IFNULL(comment, 'No comment') AS comment_display,
IFNULL(rating, 0) AS rating_display
FROM Reviews;

Output

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

Table 14: Suppliers

Question 18

SELECT supplier_id, name,
COALESCE(phone, alt_phone, 'No Contact') AS contact_number
FROM Suppliers;

Output

supplier_id	name	contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	Value Source	No Contact



Table: 15: User - Settings

Question 19

```
SELECT user_id  
       IFNULL(theme, 'Light') AS theme_set  
       IFNULL(language, 'English') AS language_set  
       IFNULL(timezone, 'UTC') AS timezone_set  
FROM User_Settings;
```

Output

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Table 16: Maintenance

Question 20

```
SELECT record_id, machine_id  
       IFNULL(issue, 'Unknown Issue') AS issue_log  
       IFNULL(technician, 'Not Assigned') AS technician_name  
FROM Maintenance;
```

Output

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex



Exercise 6

Data Analytics

Advanced SQL

DATE FUNCTIONS

Table 1: Employees

Question 1

SELECT emp_id, name, hire_date,
DATEADD(month, 6, hire_date) AS hire_plus_6_months
FROM Employees;

Output

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2: Students

Question 2

SELECT student_id, name
DATEDIFF(day, dob, CURRENT_DATE) AS age_in_days
FROM Students;

Output

student_id	name	dob	age_in_days
101	Maya	2005-08-10	7540
102	Ethan	2004-12-01	7819
103	Sienna	2006-03-15	7349



Table 3: Events

Question 3

```
SELECT event_id, event_name,  
       DATEDIFF(day, CURRENT_DATE, event_date) AS days_remaining  
FROM Events;
```

Output

event_id	event_name	event_date	days_remaining
1	Seminar	2024-06-15	-310
2	Workshop	2025-09-01	-232
3	Hackathon	2025-01-20	-455

Table 4: Invoices

Question 4

```
SELECT invoice_id, issue_date, due_date,  
       DATEDIFF(day, issue_date, due_date) AS days_between  
FROM Invoices;
```

Output

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5: Courses

Question 5

```
SELECT course_id, name,  
       TO_CHAR(start_date, 'Month YYYY') AS formatted_date  
FROM Courses;
```

Output

course_id	name	start_date	formatted_date
201	SQL Basics	2025-05-01	May 2025
202	Python	2025-06-01	June 2025



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Table 6: MembershipsQuestion 6

```
SELECT member_id,  
DATEFROMPARTS(start_year, start_month, start_day) AS quill_start_date  
FROM Memberships;
```

Output

member_id	quill_start_date
1	2023-05-10
2	2022-11-25

Table 7: SubscriptionsQuestion 7~~Output~~

```
SELECT sub_id, plan,  
DATEADD(year, 1, renewal_date) AS extended_renewal_date  
FROM Subscriptions;
```

Output

sub_id	plan	renewal_date	extended_renewal_date
11	Basic	2025-01-01	2026-01-01
12	Premium	2025-03-15	2026-03-15

Table: OrdersQuestion 8

```
SELECT order_id, order_date,  
CURRENT_DATE AS today_date,  
DATEDIFF DATEDIFF(day, order_date, CURRENT_DATE) AS days_since_order  
FROM Orders;
```

Output

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-20	371
1002	2025-04-10	2026-04-20	376



Table 9: Trainings

Question 9

```
SELECT training_id, topic,  
       EXTRACT(YEAR FROM training_date) AS training_year  
FROM Trainings;
```

Output

training_id	topic	training_date	training_year
1	Safety	2025-01-10	2025
2	Compliance	2025-02-20	2025

Table 10: Blog_Posts

Question 10

```
SELECT post_id, title,  
       DATEPART(hour, published_on) AS hour_published  
       DATEPART(minute, published_on) AS minute_published  
FROM Blog_Posts;
```

Output

post_id	title	published_on	hour_published	minute_published
1	SQL Tips	2025-04-01 10:15:00	10	15
2	Data Cleaning	2025-04-12 16:45:00	16	45

Table 11: Drivers

Question 11

```
SELECT driver_id, license_expiry,  
       DATEDIFF(day, CURRENT_DATE, license_expiry) AS days_left  
FROM Drivers;
```

Output

driver_id	license_expiry	days_left
301	2025-08-10	-254
302	2023-12-31	-841



Table 12: Messages

Question 12

```
SELECT message_id, sent_timestamp,  
CURRENT_TIMESTAMP AS current_timestamp,  
DATEDIFF(second, sent_timestamp, CURRENT_TIMESTAMP) AS seconds_since_sent  
FROM Messages;
```

Output

message_id	sent_timestamp	current_timestamp	seconds_since_sent
1	2025-04-14 09:32:45	2026-04-20 20:51:00	317 596 335
2	2025-04-18 23:59:59	2026-04-20 20:51:00	317 635 861

Table 13: Returns

Question 13

```
SELECT return_id, return_date,  
DATEADD(day, 15, return_date) AS restock_date  
FROM Returns;
```

Output

return_id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 14

Question 14

```
SELECT assign_id,  
TO_DATE(assigned_on, 'YYYY-MM-DD') AS assigned_on_date  
FROM Assignment;
```

Output

assigned_id	assigned_on	assigned_on_date
1	2025-03-01	2025-03-01
2	2025-03-05	2025-03-05



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Table 15: Meetings

Question 15

SELECT meeting_id,

TO_CHAR(scheduled_time, ('Month DD, YYYY "at" HH:MI AM')) AS formatted_meet

FROM Meetings,

meetings_id

scheduled_time

formatted_meeting_time

1

2025-04-14 14:00:00

April 14, 2025 at 02:00 PM

2

2025-04-14 09:30:00

April 14, 2025 at 09:30 AM

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```
1 SELECT * FROM `practical-big-query-493418.retail.data` LIMIT 1000;
2
3 -- 1. Filter all transactions that occurred in the year 2023 Expected Output: All columns
4 SELECT *
5 FROM `practical-big-query-493418.retail.data`
6
7
```

Query completed

Using on-demand processing quota

Query results Chat Save results Open in

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	Transaction ID	Date	Customer ID	Gender	Age
1	191	2023-10-18	CUST191	Male	
2	204	2023-09-28	CUST204	Male	

Results per page: 50 1 - 50 of 1000

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Big query practice Run Schedule Save query Download Share

```
11 -- 2. transactions where the Total Amount is more than the average Total Amount of the entire dataset
12 SELECT *
13 FROM `practical-big-query-493418.retail.data`
14 WHERE
15   `Total Amount` > (
16     SELECT AVG(`Total Amount`) FROM `practical-big-query-493418.retail.data`
17   )
```

Query completed
Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph							
Row	Transaction ID	Date	Customer ID	Gender	Age	Produ	
1	93	2023-07-14	CUST093	Female	35	Beaut	
2	74	2023-11-22	CUST074	Female	18	Beaut	

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```
--3 Calculate the total revenue (sum of Total Amount)..  
SELECT SUM(data.`Total Amount`) AS total_revenue  
FROM `practical-big-query-493418`.`retail`.`data` AS data;
```

Query completed

Using on-demand processing quota

Query results Chat Save results Open in

Job information **Results** Visualization JSON Execution details Execution graph

Row	total_revenue
1	456000

Results per page: 50 1 - 1 of 1

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```
28 --4 Display all distinct Product Categories in the dataset
29 SELECT DISTINCT data.`Product Category`
30 FROM `practical-big-query-493418`.`retail`.`data` AS data;
31
32
33
34 --5 For each Product Category calculate the total quantity sold. Expected Output: Product Category
```

Query completed

Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

Results per page: 50 1 - 3 of 3

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```
--6 Create a column called Age_Group that classifies customers as: - 'Youth' (<30) - 'Adult' (30-59) - 'Senior' (60+) Expected Output:Customer_ID, Age, Age_Group
SELECT
  data.`Customer ID`,
  data.Age,
```

Query completed

Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	Customer ID	Age	Age_Group
1	CUST191	64	Senior
2	CUST204	39	Adult

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```
--7 For each Gender, count how many high-value transactions occurred (where Total Amount > 500) Expected output
Gender, High_Value_Transactions
58 SELECT data.Gender, COUNT(data.`Transaction ID`) AS High_Value_Transactions
59 FROM `practical-big-query-493418`.`retail`.`data` AS data
60 WHERE data.`Total Amount` > 500
61 GROUP BY data.Gender;
```

Query completed

Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	Gender	High_Value_Tran...
1	Female	155
2	Male	144

Results per page: 50 1 - 2 of 2

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```
65 --8 For each Product Category, show only those categories where the total revenue exceeds 5,000.Expected Output:
66 Product_Category, Total_Revenue
67 SELECT data.`Product Category`, SUM(data.`Total Amount`) AS Total_Revenue
68 FROM `practical-big-query-493418`.`retail`.`data` AS data
69 GROUP BY data.`Product Category`
70 HAVING SUM(data.`Total Amount`) > 5000;
```

Query completed

Using on-demand processing quota

Query results Chat Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	Product Category	Total_Revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Results per page: 50 1 - 3 of 3

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```
87
88
89
90 --!0 Display all transactions from customers aged 40 or older and add a column Spending_Level showing:- 'High'
    if Total_Amount > 1000 - 'Low' otherwise Expected Output: Customer_ID, Age, Total_Amount, Spending_Level
91 SELECT
92   data.`Customer ID`
Query completed
Using on-demand processing quota
```

Query results Chat Save results Open in

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	Transaction ID	Price per Unit	Unit_Cost_Category		
1	191	25	Cheap		
2	204	25	Cheap		

Results per page: 50 1 - 50 of 1000

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```
--10 Display all transactions from customers aged 40 or older and add a column Spending_Level showing:- 'High' if Total Amount > 1000 - 'Low' otherwise Expected Output: Customer_ID, Age, Total_Amount, Spending_Level
SELECT
  data.`Customer ID`,
  data.Age,
  data.`Total Amount`,
  CASE
```

Query completed

Using on-demand processing quota

Query results Chat Save results Open in

Job information	Results	Visualization	JSON	Execution details	Execution graph
Row	Customer ID	Age	Total Amount	Spending_Level	
1	CUST191	64	25	Low	
2	CUST230	54	25	Low	

Results per page: 50 1 - 50 of 558